

Appendix A

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Table A.1 Units and Conversions

Item	English units	SI units	Conversion: English to SI
Mass	Slugs (slugs)	Kilogram (kg)	1 lb = 0.45359 kg
	Pound mass (lbm)		1 slug = 14.594 kg
	U.S. tons	Metric tonne (t)	1 U.S. ton = 0.9072 t
	Long tons		1 long ton = 1.016 t
Length	Inches (in.)	Millimeter (mm)	1 in. = 25.4 mm
	Feet (ft)	Meter (m)	1 ft = 0.3048 m
	Miles (mi)	Kilometer (km)	1 mile = 1.609 km
Area	Square feet (ft ²)	Square meter (m ²)	1 ft ² = 0.0929 m ²
Volume	Cubic inch (in. ³)	Cubic millimeter (mm ³)	1 in. ³ = 16387.0 mm ³
	Cubic feet (ft ³)	Cubic meter (m ³)	1 ft ³ = 0.02832 m ³
	U.S. gallons (gal)	Liter (L)	1 gal = 3.785 L
	Barrel (bbl)		1 bbl = 42 U.S. gal
Density	Slugs per cubic foot (slug/ft ³)	Kilogram per cubic meter (kg/m ³)	1 slug/ft ³ = 515.38 kg/m ³
Specific weight	Pound per cubic foot (lb/ft ³)	Newton per cubic meter (N/m ³)	1 lb/ft ³ = 157.09 N/m ³
Viscosity (kinematic)	Square foot per second (ft ² /s)	m ² /s	1 ft ² /s = 0.092903 m ² /s
Flow rate	Gallon/minute (gal/min)	Liter/minute (L/min)	1 gal/min = 3.7854 L/min

	Barrel/hour (bbl/hr)	Cubic meter/hour (m ³ /hr)	1 bbl/hr = 0.159 m ³ /hr
	Barrel/day (bbl/day)		
Force	Pound (lb)	Newton (N)	1 lb = 4.4482 N
Pressure	Pound/square inch (psi)	Kilopascal (kPa)	1 psi = 6.895 kPa
	Pound per square inch (lb/in. ²)	Kilogram/square centimeter (kg/cm ²)	1 psi = 0.0703 kg/cm ²
Velocity	Foot/second (ft/s)	Meter/second (m/s)	1 ft/s = 0.3048 m/s
Work and energy	British Thermal Units (Btu)	Joule (J)	1 Btu = 1055.0 J
Power	Btu/hour	Watt (W)	1 Btu/hr = 0.2931 W
		Joule/second (J/s)	
	Horsepower (HP)	Kilowatt (kW)	1 HP = 0.746 kW
Temperature	Degree Fahrenheit (°F)	Degree Celsius (°C)	1°F = 9/5°C + 32
	Degree Rankin (°R)	Degree Kelvin (K)	1°R = °F + 460
			1 K = °C + 273
Thermal conductivity	Btu/hr/ft/°F	W/m/°C	1 Btu/hr/ft/°F = 1.7307 W/m/°C
Specific heat	Btu/lb/°F	kJ/kg/°C	1 Btu/lb/°F = 4.1869 kJ/kg/°C

Table A.2 Common Properties of Petroleum Fluids

Product	Viscosity (cSt at 60°F)	API gravity	Specific gravity at 60°F	Reid vapor pressure (psi)
Regular gasoline				
Summer grade	0.70	62.0	0.7313	9.5
Interseasonal grade	0.70	63.0	0.7275	11.5
Winter grade	0.70	65.0	0.7201	13.5
Premium gasoline				
Summer grade	0.70	57.0	0.7467	9.5
Interseasonal grade	0.70	58.0	0.7165	11.5
Winter grade	0.70	66.0	0.7711	13.5
No. 1 fuel oil	2.57	42.0	0.8155	
No. 2 fuel oil	3.90	37.0	0.8392	
Kerosene	2.17	50.0	0.7796	
Jet fuel JP-4	1.40	52.0	0.7711	2.7
Jet fuel JP-5	2.17	44.5	0.8040	

Table A.3 Specific Gravity and API Gravity

Liquid	Specific gravity at 60°F	API gravity at 60°F
Propane	0.5118	N/A
Butane	0.5908	N/A
Gasoline	0.7272	63.0
Kerosene	0.7796	50.0
Diesel	0.8398	37.0
Light crude	0.8348	38.0
Heavy crude	0.8927	27.0
Very heavy crude	0.9218	22.0
Water	1.0000	10.0

Table A.4 Viscosity Conversions

Viscosity (SSU)	Viscosity (cSt)	Viscosity (SSF)
31.0	1.00	
35.0	2.56	
40.0	4.30	
50.0	7.40	
60.0	10.30	
70.0	13.10	12.95
80.0	15.70	13.70
90.0	18.20	14.44
100.0	20.60	15.24
150.0	32.10	19.30
200.0	43.20	23.50
250.0	54.00	28.0
300.0	65.00	32.5
400.0	87.60	41.9
500.0	110.00	51.6
600.0	132.00	61.4
700.0	154.00	71.1
800.0	176.00	81.0
900.0	198.00	91.0
1000.0	220.00	100.7
1500.0	330.00	150
2000.0	440.00	200
2500.0	550.00	250
3000.0	660.00	300
4000.0	880.00	400
5000.0	1100.00	500
6000.0	1320.00	600
7000.0	1540.00	700
8000.0	1760.00	800
9000.0	1980.00	900
10000.0	2200.00	1000
15000.0	3300.00	1500
20000.0	4400.00	2000

Table A.5 Thermal Conductivities

Substance	Thermal conductivity (Btu/hr/ft/°F)
Fire clay brick (burnt at 2426°F)	0.60 to 0.63
Fire clay brick (burnt at 2642°F)	0.74 to 0.81
Fire clay brick (Missouri)	0.58 to 1.02
Portland cement	0.17
Mortar cement	0.67
Concrete	0.47 to 0.81
Cinder concrete	0.44
Glass	0.44
Granite	1.0 to 2.3
Limestone	0.73 to 0.77
Marble	1.6
Sandstone	0.94 to 1.2
Corkboard	0.025
Fiber insulating board	0.028
Aerogel, silica	0.013
Coal, anthracite	0.15
Coal, powdered	0.067
Ice	1.28
Sandy soil, dry	0.25 to 0.40
Sandy soil, moist	0.50 to 0.60
Sandy soil, soaked	1.10 to 1.30
Clay soil, dry	0.20 to 0.30
Clay soil, moist	0.40 to 0.50
Clay soil, moist to wet	0.60 to 0.90
River water	2.00 to 2.50
Air	2.00

Table A.6 Absolute Roughness of Pipe

Pipe material	Roughness (mm)	Roughness (in.)
Riveted steel	0.9 to 9.0	0.0354 to 0.354
Concrete	0.3 to 3.0	0.0118 to 0.118
Wood stave	0.18 to 0.9	0.0071 to 0.0354
Cast iron	0.26	0.0102
Galvanized iron	0.15	0.0059

(Continued)

Table A.6 Continued

Pipe material	Roughness (mm)	Roughness (in.)
Asphalted cast iron	0.12	0.0047
Commercial steel	0.045	0.0018
Wrought iron	0.045	0.0018
Drawn tubing	0.0015	0.000059

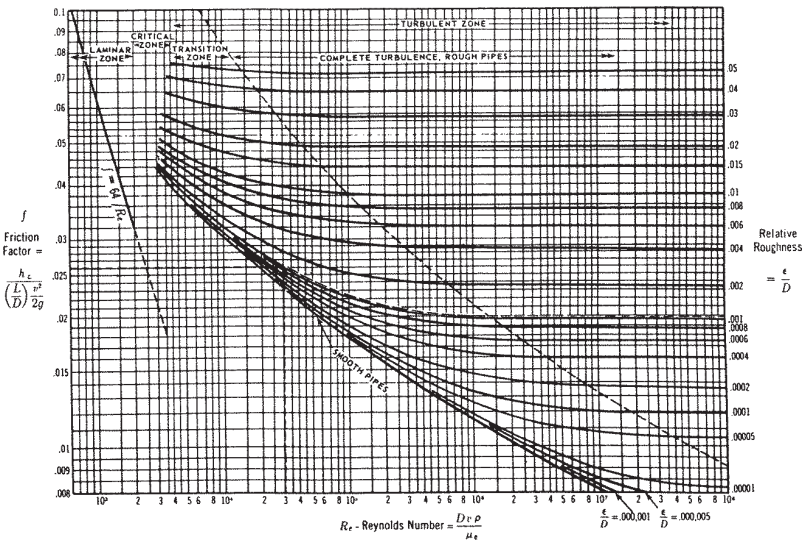


Figure A.7 Moody diagram.

Table A.8 Typical Hazen-Williams C-Factors

Pipe material	C-factor
Smooth pipes (all metals)	130–140
Smooth wood	120
Smooth masonry	120
Vitrified clay	110
Cast iron (old)	100
Iron (worn/pitted)	60–80
Polyvinyl chloride (PVC)	150
Brick	100

Table A.9 Friction Loss in Valves: Resistance Coefficient K

Description	L/D	Nominal pipe size (in.)											
		1/2	3/4	1.0	1¼	1½	2	2½ to 3	4	6	8 to 10	12 to 16	18 to 24
Gate valve	8	0.22	0.20	0.18	0.18	0.15	0.15	0.14	0.14	0.12	0.11	0.10	0.10
Globe valve	340	9.2	8.5	7.8	7.5	7.1	6.5	6.1	5.8	5.1	4.8	4.4	4.1
Ball valve	3	0.08	0.08	0.07	0.07	0.06	0.06	0.05	0.05	0.05	0.04	0.04	0.04
Butterfly valve							0.86	0.81	0.77	0.68	0.63	0.35	0.30
Plug valve, straightway	18	0.49	0.45	0.41	0.40	0.38	0.34	0.32	0.31	0.27	0.25	0.23	0.22
Plug valve, three-way thru-flo	30	0.81	0.75	0.69	0.66	0.63	0.57	0.54	0.51	0.45	0.42	0.39	0.36
Plug valve, branch-flo	90	2.43	2.25	2.07	1.98	1.89	1.71	1.62	1.53	1.35	1.26	1.17	1.08

Table A.10 Equivalent Lengths of Valves and Fittings

Description	L/D
Gate valve	8
Globe valve	340
Ball valve	3
Swing check valve	50
Standard elbow, 90°	30
Standard elbow, 45°	16
Long-radius elbow, 90°	16

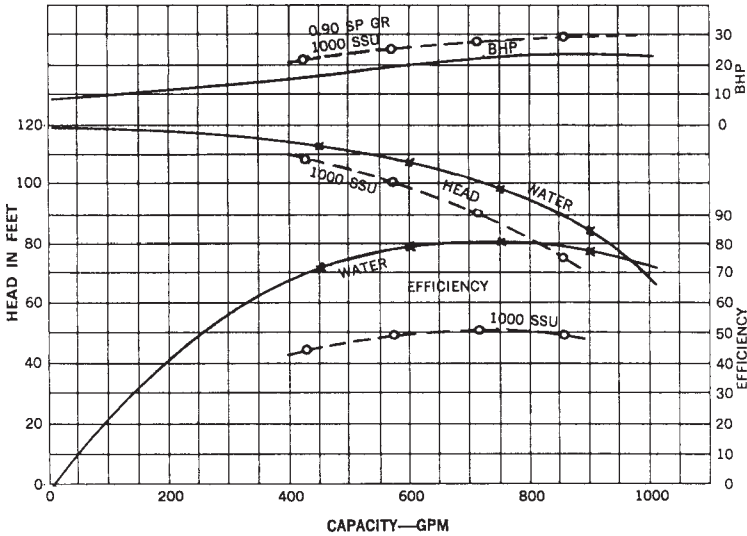
Example: A 14-in. gate valve has L/D ratio=8. Therefore equivalent length=8×14 in.=112 in.=9.25 feet.

Table A.11 Seam Joint Factors for Pipes

Specification	Pipe class	Seam joint factor (E)
ASTM A53	Seamless	1.00
	Electric resistance welded	1.00
	Furnace lap welded	0.80
	Furnace butt welded	0.60
ASTM A106	Seamless	1.00
ASTM A134	Electric fusion arc welded	0.80
ASTM A135	Electric resistance welded	1.00
ASTM A139	Electric fusion welded	0.80
ASTM A211	Spiral welded	0.80
ASTM A333	Seamless	1.00
	Welded	1.00
ASTM A381	Double submerged arc welded	1.00
ASTM A671	Electric fusion welded	1.00
ASTM A672	Electric fusion welded	1.00
ASTM A691	Electric fusion welded	1.00
API 5L	Seamless	1.00
	Electric resistance welded	1.00
	Electric flash welded	1.00
	Submerged arc welded	1.00
	Furnace lap welded	0.80
	Furnace butt welded	0.60
API 5LX	Seamless	1.00
	Electric resistance welded	1.00
	Electric flash welded	1.00
	Submerged arc welded	1.00
API 5LS	Electric resistance welded	1.00
	Submerged arc welded	1.00

Table A.12 ANSI Pressure Ratings

Class	Allowable pressure (psi)
150	275
300	720
400	960
600	1440
900	2160
1500	3600



SAMPLE CALCULATIONS				
	0.6 x Q _{NW}	0.8 x Q _{NW}	1.0 x Q _{NW}	1.2 x Q _{NW}
Water capacity (Q _w)	450	600	750	900
Water head in feet (H _w)	114	108	100	86
Water efficiency (E _w)	72.5	80	82	79.5
Viscosity of liquid	1000 SSU	1000 SSU	1000 SSU	1000 SSU
C _Q —from chart	0.95	0.95	0.95	0.95
C _H —from chart	0.96	0.94	0.92	0.89
C _E —from chart	0.635	0.635	0.635	0.635
Viscous capacity—Q _w x C _Q	427	570	712	855
Viscous head—H _w x C _H	109.5	101.5	92	76.5
Viscous efficiency—E _w x C _E	46.0	50.8	52.1	50.5
Specific gravity of liquid	0.90	0.90	0.90	0.90
bhp viscous	23.1	25.9	28.6	29.4

Figure A.13 Centrifugal pump performance viscosity correction. (Courtesy of Hydraulic Institute, Parsippany, NJ; www.pumps.org.)

Table A.14 Approximate Pipeline Construction Cost

Pipe diameter (in.)	Average cost (\$/in.-dia/mile)
8	18,000
10	20,000
12	22,000
16	14,900
20	20,100
24	33,950
30	34,600
36	40,750

Table A.15 Thermal Hydraulics Summary

***** TEMPERATURE AND PRESSURE PROFILE *****							
Distance (mi)	FlowRate (bbl/day)	Temperature (degF)	SpGrav --	Viscosity CST	Pressure (psi)	MAOP (psi)	Location
0.00	100,000.00	190.00	0.9355	68.72	50.00	1400.00	Compton
0.00	100,000.00	192.74	0.9345	63.93	1181.18	1400.00	Compton
18.00	100,000.00	186.76	0.9366	74.96	792.70	1400.00	
22.00	100,000.00	185.50	0.9371	77.59	298.41	1400.00	
23.00	100,000.00	185.19	0.9372	78.26	299.60	1400.00	
24.00	100,000.00	184.88	0.9373	78.93	300.74	1400.00	
29.00	100,000.00	183.35	0.9378	82.36	306.20	1400.00	
35.00	100,000.00	181.56	0.9385	86.60	248.57	1400.00	
40.00	100,000.00	180.12	0.9390	90.24	50.00	1400.00	Dimpton
40.00	100,000.00	183.34	0.9378	82.39	1371.78	1400.00	Dimpton
40.58	100,000.00	183.16	0.9379	82.79	1348.95	1400.00	
50.00	100,000.00	180.38	0.9389	89.56	977.68	1400.00	
54.00	100,000.00	179.25	0.9393	92.52	868.04	1400.00	
55.00	100,000.00	178.97	0.9394	93.28	865.82	1400.00	
60.00	100,000.00	177.58	0.9398	97.10	813.82	1400.00	
70.00	100,000.00	174.91	0.9408	105.06	666.49	1400.00	
80.00	100,000.00	172.37	0.9417	113.38	391.64	1400.00	
90.00	100,000.00	169.97	0.9425	122.06	213.27	1400.00	
100.00	100,000.00	167.69	0.9433	131.08	50.00	1400.00	Terminus

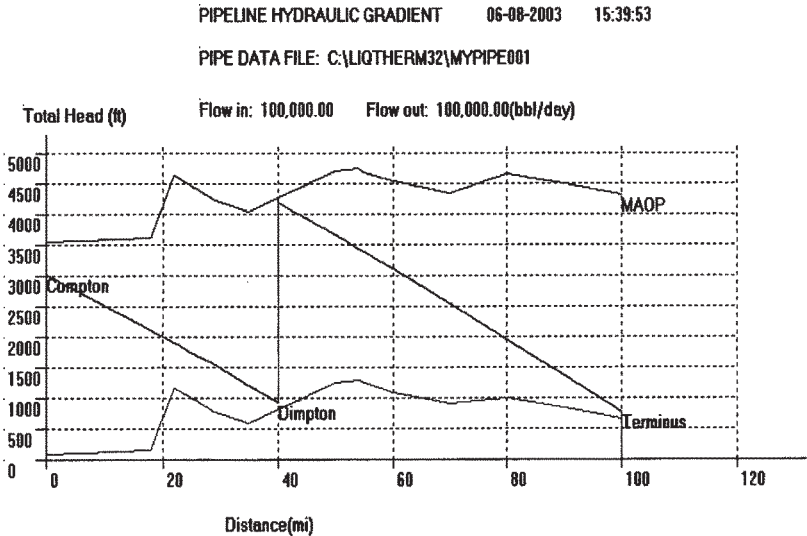


Figure A.16 Thermal hydraulics pressure gradient.

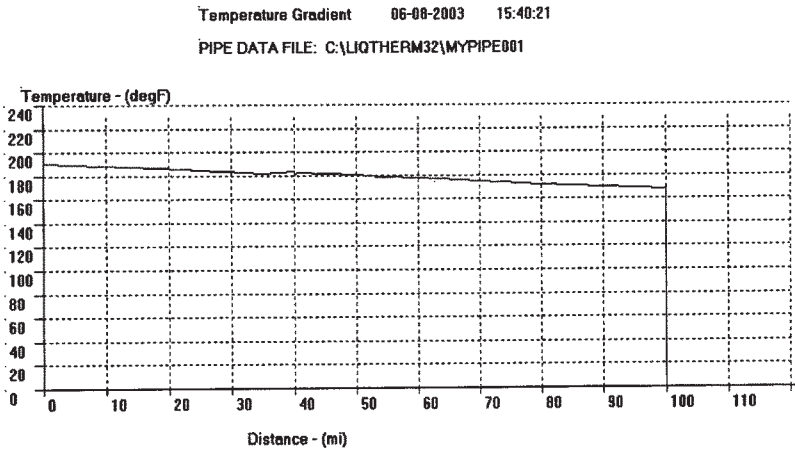


Figure A.17 Thermal hydraulics temperature gradient.